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Comprehensive voltage-loss analysis and reduction of radiative recombination voltage loss in quantum-structured solar cells

Meita Asami [a,*], Maui Hino [b], Gan Li [b], Kentaroh Watanabe [a], Yoshiaki Nakano [b], Masakazu Sugiyama [a,b]

[a] Research Center for Advanced Science and Technology, The University of Tokyo, 4-6-1 Komaba, Meguro, Tokyo, 153-8904, Japan
[b] School of Engineering, The University of Tokyo, 7-3-1 Hongo, Bunkyo, Tokyo, 113-8654, Japan

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ABSTRACT

Voltage-loss analysis is essential in the development of next-generation solar cells, such as perovskite, chalcopyrite, kesterite, and nano/quantum-structured solar cells. Voltage-loss analysis provides valuable insights into how the energy conversion efficiency of solar cells can be enhanced. However, a comprehensive and accurate method to evaluate the voltage loss in quantum-structured solar cells is lacking. This study establishes and demonstrates a quantitative voltage-loss analysis based on detailed balance theory. This analysis reveals the relationship between external quantum efficiency and radiative recombination voltage loss in quantum-structured solar cells. Based on the results of the analysis, we designed and fabricated a novel low-voltage loss quantum-structured solar cell. Radiative recombination in the quantum-structured solar cell was successfully suppressed by steepening the absorption edge. This voltage-loss analysis facilitates the development of next-generation solar cells.

1. Introduction

For humanity to achieve a sustainable society, the reliance on fossil fuels should be reduced progressively, and conventional primary energy sources should be replaced with renewable energy sources. The sun beams enormous amounts of energy to the Earth, and harnessing solar power effectively is essential for establishing a sustainable decarbonized energy system. A key requirement for the realization of a sustainable society is the decarbonization of vehicles, such as cars [1]. To address this, solar cars and vehicles equipped with solar cells and batteries have been developed. However, the efficiency of solar cells must be improved to increase the cruising range of solar cars without charging [2]. By achieving a longer cruising range, drivers can enjoy the convenience and comfort of long-distance travel without the hassle of frequent stops at electric charging stations. Therefore, solar cells with high energy-conversion efficiencies are in demand, and III-V compound solar cells are attracting significant attention.

Studies on the application of III-V compound solar cells are underway. Inverted metamorphic (IMM) triple-junction solar cells, which are a type of III-V compound multi-junction solar cells, have been integrated into the Toyota Prius plug-in hybrid vehicle [2]. Conventional IMM triple-junction solar cells consist of InGaP/GaAs/InGaAs. The energy in a wide range of sunlight spectra can be efficiently converted to electrical energy by combining multiple semiconductor materials with different bandgaps. Light is irradiated to the top side of the InGaP subcell; higher-energy photons are absorbed in a wider bandgap material, lower-energy photons are absorbed in GaAs, and even lower-energy photons are absorbed in InGaAs. III-V semiconductor multijunction solar cells can reduce the thermalization and transmission losses, which are the primary sources of energy loss in solar cells, and achieve higher energy conversion efficiency compared with single junction solar cells.

For the practical terrestrial utilization of III-V solar cells, we must reduce the fabrication costs of III-V semiconductor multi-junction solar cells [3]. Substrate reuse and enhancement of energy conversion efficiency are the major factors for $/W cost reduction. The epitaxial lift-off technique has been investigated for substrate reuse; however, this is beyond the scope of this study [4,5]. This study focuses on strategies to improve the energy conversion efficiency of conventional triple-junction solar cells. Conventional triple-junction solar cells face a problem with the bandgap of GaAs middle cells. The bandgap of GaAs is excessively large, resulting in low current generation in the middle cell. To increase the energy conversion efficiency, the bandgap of the middle cell should

* Corresponding author.
E-mail address: meita-asami@g.ecc.u-tokyo.ac.jp (M. Asami).

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be reduced and the number of photo-generated carriers should be increased. Previous studies adjusted the GaAs bandgap by inserting an InGaAs/GaAsP strain-balanced quantum structure into a GaAs solar cell [6–10]. However, although the insertion of quantum structures can increase the current, the voltage is lower than that of GaAs solar cells [6–8,10,11].

In this study, we quantitatively analyze the causes of voltage drop in GaAs solar cells when quantum structures are inserted. The voltage drop is a sum of the voltage loss and inevitable voltage drop owing to the insertion of a material with a narrow bandgap; however, to date, an accurate evaluation metric is absent. We propose an evaluation method for voltage loss based on the detailed balance theory and demonstrate its suitability. We designed and fabricated new quantum structures with reduced voltage loss based on the knowledge obtained from the analysis of voltage loss using the proposed method. The proposed voltage-loss analysis is applicable to all quantum/nanostructured solar cells, as well as to perovskite, chalcopyrite, and kesterite solar cells. This voltage-loss analysis is expected to accelerate the development of next-generation solar cells.

2. Theory

This section describes the voltage-loss analysis method employed in this study. Voltage losses in solar cells can be determined using the widely recognized detailed balance theory [12,13]. V_{OC}^{SQ} is the open-circuit voltage at the Shockley–Queisser limit, calculated using the detailed balance theory. V_{OC}^{SQ} represents the theoretical limit/ideal voltage of a solar cell with no voltage losses constructed from a material with a certain bandgap, E_g. V_{OC}^{SQ} is calculated as

$$V_{OC}^{SQ} = \frac{kT}{q}\ln\frac{J_{SC}^{SQ}}{J_{em,0}^{SQ}} = \frac{kT}{q}\ln\frac{q\int_0^\infty \varphi_{sun} u_{EQE}(E)\,dE}{q\int_0^\infty \varphi_{bb} u_{EQE}(E)\,dE} = \frac{kT}{q}\ln\frac{q\int_{E_g}^\infty \varphi_{sun}\times 1\,dE}{q\int_{E_g}^\infty \varphi_{bb}\times 1\,dE}, \quad (1)$$

where k is the Boltzmann constant and T is the temperature. J_{SC}^{SQ} and $J_{em,0}^{SQ}$ are the short-circuit current density and diode saturation current density at the Shockley-Queisser limit, respectively, given by Eqs. (2) and (3). φ_{sun} and φ_{bb} are the photon fluxes of sunlight and black body radiation, respectively. $u_{EQE}(E)$ is the step-function external quantum efficiency (EQE), given by Eq. (4).

$$J_{SC}^{SQ} = q\int_0^\infty \varphi_{sun} u_{EQE}(E) dE, \quad (2)$$

$$J_{em,0}^{SQ} = q\int_0^\infty \varphi_{bb} u_{EQE}(E)\,dE, \quad (3)$$

$$u_{EQE}(E) = \begin{cases} 1 & E > E_g \\ 0 & E \le E_g \end{cases}. \quad (4)$$

The voltage loss ΔV is evaluated as

$$\Delta V = V_{OC}^{SQ} - V_{OC}, \quad (5)$$

where V_{OC} is the open-circuit voltage of the solar cell. ΔV can be divided into three components: radiative recombination voltage loss ΔV_{OC}^{rad}, nonradiative recombination voltage loss ΔV_{OC}^{nonrad}, and short-circuit current voltage loss ΔV_{OC}^{SC}. Each voltage loss is characterized as follows:

$$\Delta V = V_{OC}^{SQ} - V_{OC} = \frac{kT}{q}\ln\frac{J_{SC}^{SQ}}{J_{em,0}^{SQ}} - \frac{kT}{q}\ln\frac{J_{SC}-J_{nr}}{J_{em,0}}$$
$$= -\frac{kT}{q}\ln\frac{J_{SC}}{J_{SC}^{SQ}}\times\frac{J_{em,0}^{SQ}}{J_{em,0}}\times\frac{J_{SC}-J_{nr}}{J_{SC}} \quad (6)$$

$$= \left(-\frac{kT}{q}\ln\frac{J_{em,0}^{SQ}}{J_{em,0}}\right) + \left(-\frac{kT}{q}\ln\frac{J_{SC}-J_{nr}}{J_{SC}}\right) + \left(-\frac{kT}{q}\ln\frac{J_{SC}}{J_{SC}^{SQ}}\right) = \Delta V_{OC}^{rad} + \Delta V_{OC}^{nonrad} + \Delta V_{OC}^{SC}, \quad (7)$$

$$\Delta V_{OC}^{rad} = -\frac{kT}{q}\ln\frac{J_{em,0}^{SQ}}{J_{em,0}}, \quad (8)$$

$$\Delta V_{OC}^{nonrad} = -\frac{kT}{q}\ln\frac{J_{SC}-J_{nr}}{J_{SC}} = \frac{kT}{q}ln\left(\frac{J_{SC}}{J_{em,0}}\right) - V_{OC}, \quad (9)$$

$$\Delta V_{OC}^{SC} = -\frac{kT}{q}\ln\frac{J_{SC}}{J_{SC}^{SQ}}, \quad (10)$$

where J_{SC} is the actual short-circuit current density of the solar cell, $J_{em,0}$ is the actual diode saturation current density of the solar cell, and J_{nr} is the nonradiative recombination current. J_{SC} and $J_{em,0}$ can be expressed as follows:

$$J_{SC} = q\int_0^\infty \varphi_{sun} Q_e(E) dE, \quad (11)$$

$$J_{em,0} = q\int_0^\infty \varphi_{bb} Q_e(E) dE, \quad (12)$$

where Q_e is the actual EQE of the solar cell. Conventionally, the voltage losses are calculated using Eqs. (1)–(5). In this study, we used the following step function instead of Eq. (4):

$$u_{EQE_{max}}(E) = \begin{cases} EQE_{max} & E > E_g \\ 0 & E \le E_g \end{cases}, \quad (13)$$

where EQE_{max} denotes the maximum EQE. A conventional method [12] using Eq. (4) cannot accurately evaluate the voltage loss derived from materials in certain situations. For example, the conventional method falls short when comparing samples with and without an antireflection coating on a solar cell. The sample with the antireflection coating exhibits a smaller short-circuit current voltage loss; however, this is not attributable to the quality of the solar cell material, rather to the surface treatment. Additionally, the conventional method fails to accurately assess voltage loss when the light-absorption layer is thin. In such cases, the voltage loss often becomes negative, which is physically unachievable. Consequently, the conventional method is not applicable to those types of solar cells. In this study, we employed the EQE in Eq. (13) to eliminate the effects of surface reflectance and thickness of the light-absorption layers. Each type of voltage loss can be expressed as follows:

$$\Delta V = V_{OC}^{SQ} - V_{OC} = \frac{kT}{q}\ln\frac{J_{SC}^{SQ}}{J_{em,0}^{SQ}} - \frac{kT}{q}\ln\frac{J_{SC}-J_{nr}}{J_{em,0}}$$
$$= \frac{kT}{q}\ln\frac{EQE_{max}\,J_{SC}^{SQ}}{EQE_{max}\,J_{em,0}^{SQ}} - \frac{kT}{q}\ln\frac{J_{SC}-J_{nr}}{J_{em,0}}$$

$$= -\frac{kT}{q}\ln\frac{J_{SC}}{J_{SC}^{SQmax}}\times\frac{J_{em,0}^{SQmax}}{J_{em,0}}\times\frac{J_{SC}-J_{nr}}{J_{SC}}$$

$$= \left(-\frac{kT}{q}\ln\frac{J_{em,0}^{SQmax}}{J_{em,0}}\right) + \left(-\frac{kT}{q}\ln\frac{J_{SC}-J_{nr}}{J_{SC}}\right) + \left(-\frac{kT}{q}\ln\frac{J_{SC}}{J_{SC}^{SQmax}}\right) \equiv \Delta V_{OC}^{rad} + \Delta V_{OC}^{nonrad} + \Delta V_{OC}^{SC}, \quad (14)$$

$$\Delta V_{OC}^{rad} \equiv -\frac{kT}{q}\ln\frac{J_{em,0}^{SQmax}}{J_{em,0}}, \quad (8)$$

$$\Delta V_{OC}^{nonrad} = -\frac{kT}{q}\ln\frac{J_{SC}-J_{nr}}{J_{SC}} = \frac{kT}{q}ln\left(\frac{J_{SC}}{J_{em,0}}\right) - V_{OC}, \tag{9}$$

$$\Delta V_{OC}^{SC} \equiv -\frac{kT}{q}\ln\frac{J_{SC}}{J_{SC}^{SQmax}}, \tag{10}$$

where

$$J_{SC}^{SQmax} = EQE_{max}\,J_{SC}^{SQ} = q\int_0^\infty \varphi_{sun}u_{EQE_{max}}(E)dE, \tag{15}$$

$$J_{em,0}^{SQmax} = EQE_{max}\,J_{em,0}^{SQ} = q\int_0^\infty \varphi_{bb}u_{EQE_{max}}(E)dE. \tag{16}$$

The definition of the total voltage loss, $\Delta V = V_{OC}^{SQ} - V_{OC}$, is the same as in the conventional method. However, we redefined ΔV_{OC}^{rad} and ΔV_{OC}^{SC} to provide a more generalized voltage-loss analysis.

Nonradiative recombination voltage loss ΔV_{OC}^{nonrad} is determined using nonradiative recombination current J_{nr}, as presented in Eq. (9). Notably, the optoelectronic reciprocity relationship holds between nonradiative recombination and voltage loss [12]. Crystal defects induce nonradiative recombination, resulting in the dissipation of electrical energy as heat.

Radiative recombination voltage loss ΔV_{OC}^{rad} is caused by excess expansion of absorption edge toward lower energy levels (longer wavelength). We can understand the radiative recombination voltage loss as the additional voltage loss that is added to the well-known inevitable emission loss derived from Kirchhoff's law [13]. In reality, a solar cell cannot achieve a step-like absorption edge as presented in Eqs. (4) and (13), and shown in Fig. 1(a). Thus, an absorption tail exists, as indicated by the black dotted circle in Fig. 1(a). This tail increases the diode saturation current; for example, we calculated the diode saturation current from EQEs, EQE_{SQ}, and EQE_{actual}, and obtained $J_{em,0}^{SQmax} = 3.57 \times 10^{-18}$ mA/cm^2 and $J_{em,0} = 4.67 \times 10^{-17}$ mA/cm^2. The dark currents are presented in Fig. 1(b). The difference between $J_{em,0}^{SQmax}$ and $J_{em,0}$ results in the radiative recombination voltage loss, $\Delta V_{OC}^{rad} = -\frac{kT}{q}\ln\frac{J_{em,0}^{SQmax}}{J_{em,0}}$. The method employed in defining the bandgap and EQE_{SQ} to calculate $J_{em,0}^{SQmax}$ is described hereafter (see Fig. 3).

The short-circuit current voltage loss results from the lower short-circuit current density of the actual solar cell. Generally, the difference between J_{SC}^{SQ} and J_{SC} is insignificant, and ΔV_{OC}^{SC} is approximately a few meV. Compared with ΔV_{OC}^{nonrad} and ΔV_{OC}^{rad}, ΔV_{OC}^{SC} is considered negligible. When we calculate the voltage losses, we must consider the EQE measurement noise. The noise is observed in areas where the EQE value is close to zero, as shown in Fig. 2. The presence of noise near zero EQE at longer wavelengths significantly influences V_{OC}^{SQ}, and considerably degrades the accuracy of the voltage-loss analysis. Thus, the noise

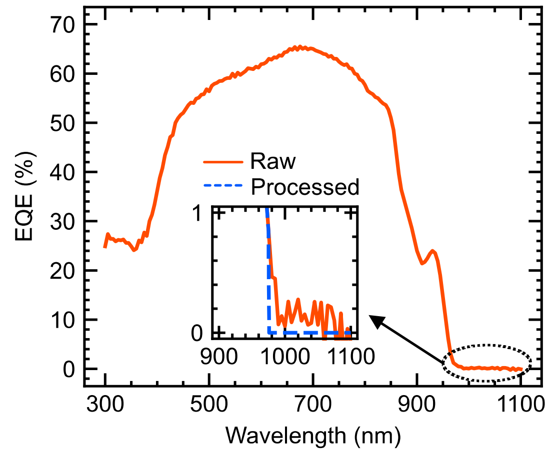

Fig. 2. Data processing for measurement noise at the low value of EQE (inset; magnified EQE near zero).

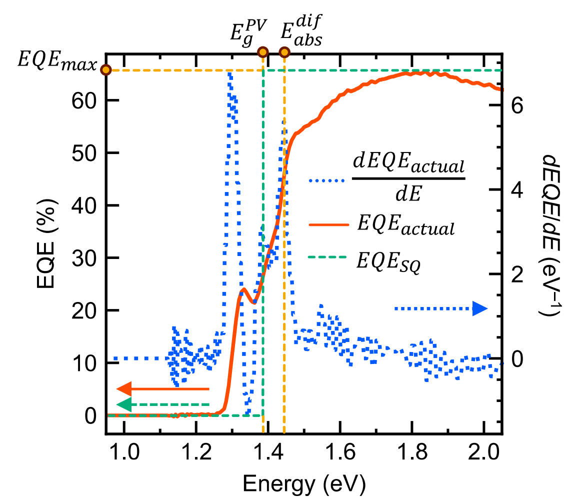

Fig. 3. EQE of an actual solar cell and a solar cell with an ideal absorption edge (primary vertical axis) and the first derivative of actual EQE (secondary vertical axis).

treatment should be implemented according to the noise level of the EQE measurements, as shown in Fig. 2. In this study, we considered an EQE of 0.8 % or less as noise.

The bandgap of the entire solar cell must be determined when performing voltage-loss analysis using the aforementioned equations. Next-generation solar cells often incorporate various semiconductor

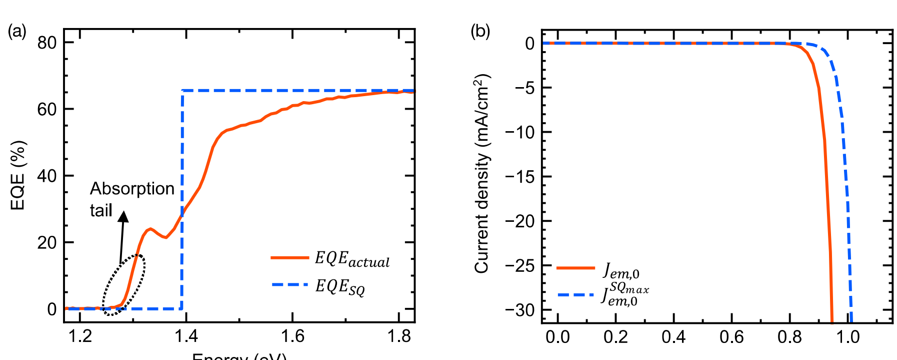

Fig. 1. (a) External quantum efficiency, and (b) dark current density of an actual solar cell (red line) and a solar cell with step EQE (blue dashed line).

materials. InGaP/GaAs heterojunction solar cells use InGaP as an emitter layer and GaAs as a base layer. Heterojunction solar cells have been investigated to suppress nonradiative recombination [14,15]. Another example is the quantum-structured solar cells, which consists of several materials, such as InGaAs/GaAsP. However, determining the bandgaps of such solar cells is challenging. For example, in GaAs/InGaP heterojunction solar cells, the bandgap of GaAs is 1.42 eV and that of InGaP is 1.86 eV. Consequently, defining the bandgap for the entire solar cell poses a significant challenge. The complexity of quantum-structure solar cells increases owing to the intricate combination of multiple materials, thicknesses, and compositions.

To date, no method has been proposed to logically and rationally determine the bandgap of an entire complex solar cell. Several methods have been employed to determine the bandgap of a solar cell [12,16–20]; however, they are unsuitable for samples with complex shapes of EQEs, such as quantum-structured solar cells [11]. Some of these methods use the peak of the first derivative of the EQE to define the bandgap, which corresponds to the absorption edge [12,20]. However, relying solely on the peak of the first derivative of the EQE is insufficient for calculating the bandgap required for voltage-loss analysis, because some solar cells with complex structure have several peaks as shown in Fig. 3.

In this study, we propose a novel method to determine the bandgap of solar cells. Our approach goes beyond solely focusing on the absorption edge, as determined from the first derivative of the EQE, and considers the photocurrent density. The underlying motivation for reducing the bandgap of solar cells is to increase the photocurrent density. By using materials with a narrower bandgap, solar cells can effectively convert lower-energy photons into carriers and consequently achieve higher current densities, albeit with a reduced voltage. This trade-off between current density and voltage is fundamental to solar cells. To achieve optimal energy conversion efficiency, the bandgap and voltage should be maximized while maintaining a certain current density. This maximized bandgap can then be used to calculate the ideal voltage V_{OC}^{SQ}.

The proposed method for calculating the bandgap of solar cells is as follows. First, the first derivative of EQE is calculated as shown in Fig. 3. The peak of the first derivative on the higher-energy (shorter-wavelength) side is identified as the absorption edge E_{abs}^{dif}. The current J_{ex} generated by the EQE on the low-energy side below the absorption edge is calculated as

$$J_{ex} = q\int_0^{E_{abs}^{dif}} \varphi_{sun} Q_e dE. \tag{17}$$

Then, the bandgap of the entire solar cell, E_g^{PV}, is calculated numerically as

$$q\int_{E_g^{PV}}^{E_{abs}^{dif}} \varphi_{sun} EQE_{max} dE = q\int_0^{E_{abs}^{dif}} \varphi_{sun} Q_e dE = J_{ex}. \tag{18}$$

E_g^{PV} is shown in Fig. 3. In Eq. (18), the maximum value of the EQE is used to evaluate the intrinsic quality of the semiconductor material. We can eliminate the influence of antireflection coatings and light absorbance, the latter depends on the thickness of the light-absorption layer, by using EQE_{max}. The E_g^{PV} value, determined through this calculation, is defined as the ideal bandgap of the solar cell in this study, and the voltage loss is calculated using this bandgap and Eq. (13).

3. Experimental

We employed the metal-organic vapor phase epitaxy (MOVPE) technique with the AIX200/4 (AIXTRON) reactor to grow quantum-structured solar cell samples. The main precursors employed in the MOVPE were trimethylgallium (TMGa), trimethylindium (TMIn), tertiary butylarsine (TBA), and tertiary butylphosphine (TBP), and the dopant precursors were dimethylzinc (DMZn) and hydrogen sulfide. Triethylgallium (TEGa) was used as a gallium precursor for fabricating the InGaAs/GaAs/GaAsP quantum structure. The carrier gas was hydrogen, and the reactor pressure was 100 mbar.

The structures of the solar cell samples are shown in Fig. 4. Six types of solar cells were fabricated. The first was a GaAs solar cell without a quantum structure (GaAs ref.), as shown in Fig. 4(a). The two samples had planar superlattices (PSL) of 20 and 50 repeats (20PSL and 50PSL). The other two samples had wire-on-well superlattices (WoW) of 20 and 50 repeats (20 WoW and 50 WoW). The PSL and WoW solar cells are shown in Fig. 4(b). Cross-sectional STEM images of planar and WoW superlattices are available in Ref. [21]. Finally, we developed a sample with thick well planar superlattices (TW-PSL). The schematics of each quantum structure are shown in Fig. 4(b)' and (c)'. The quantum structures were strain-balanced; the lattice constant of InGaAs is larger than that of GaAs and the lattice constant of GaAsP is smaller than that of GaAs. By growing InGaAs and GaAsP thinly and alternately, we could compensate and balance the crystal strain.

After crystal growth, a back electrode (~450-nm Au on 5-nm Ag) was deposited onto the sample using a thermal evaporator. The top electrode (300-nm Au on 200-nm AuGe) was deposited using an electron beam evaporator. Thermal annealing was not performed on the samples. The cell sizes of the GaAs ref., 20PSL, 50PSL, 20WoW, and 50WoW were 2 mm × 2 mm, and the aperture area was 3.01 mm^2. The cell size of TW-PSL was 5.5 mm × 5.5 mm, and the aperture area was 25.6 mm^2. Antireflection coating was not applied to the samples.

WoW is a superlattice, which forms spontaneously when a superlattice is grown on a vicinal substrate at an appropriate temperature [22]. Step bunching forms several crystal terraces that become WoW structures. WoW has higher carrier mobility and longer carrier lifetime compared with PSL [23]. In this study, GaAs ref., 20WoW, 50WoW, and TW-PSL were grown on misoriented GaAs (0 0 1) substrates (6° off toward <1 1 1>B). Because the InGaAs quantum-well layers of the TW-PSL were quite thick, undulation did not occur, and a planar superlattice was formed despite the use of a substrate with misorientation identical to that of the WoW samples.

4. Results and discussion

We conducted a comprehensive analysis of voltage loss in quantum-structured solar cells. We measured their current–voltage (*J*–*V*) characteristics and EQEs, as shown in Fig. 5(a) and (b), and Table 1. Notably, the samples containing PSL exhibited a lower carrier collection efficiency, resulting in reduced current density and EQE. The voltage was lower for the WoW samples. This finding aligns with a previous study in which it was reported that WoW samples had lower voltage compared with PSL samples, although the underlying reason was not clarified [21]. To investigate this phenomenon, we employed the method described in Section 2, and analyzed the voltage loss in quantum-structured solar cells in terms of their *J*–*V* characteristics and EQEs. We aimed to understand the reasons for the lower voltage in the WoW samples compared with PSL samples. The results of this analysis are shown in Fig. 6. The sample with the quantum structure showed a lower nonradiative recombination voltage loss compared with that of the GaAs ref. Sample. This can be attributed to the confinement of carriers within narrow quantum wells in the quantum structure. This confinement enhances the probability of radiative recombination, effectively suppressing nonradiative recombination.

The WoW sample exhibited a larger nonradiative recombination voltage loss compared with the PSL samples, as shown in Fig. 6. This is believed to have resulted from the undulated quantum-well layer in WoW, which increased the area of the InGaAs/GaAsP heterointerfaces and the number of crystal defects generated at the heterointerfaces. A more detailed analysis of the nonradiative recombination voltage loss is presented in our previous study [11]. The voltage-loss analysis results of

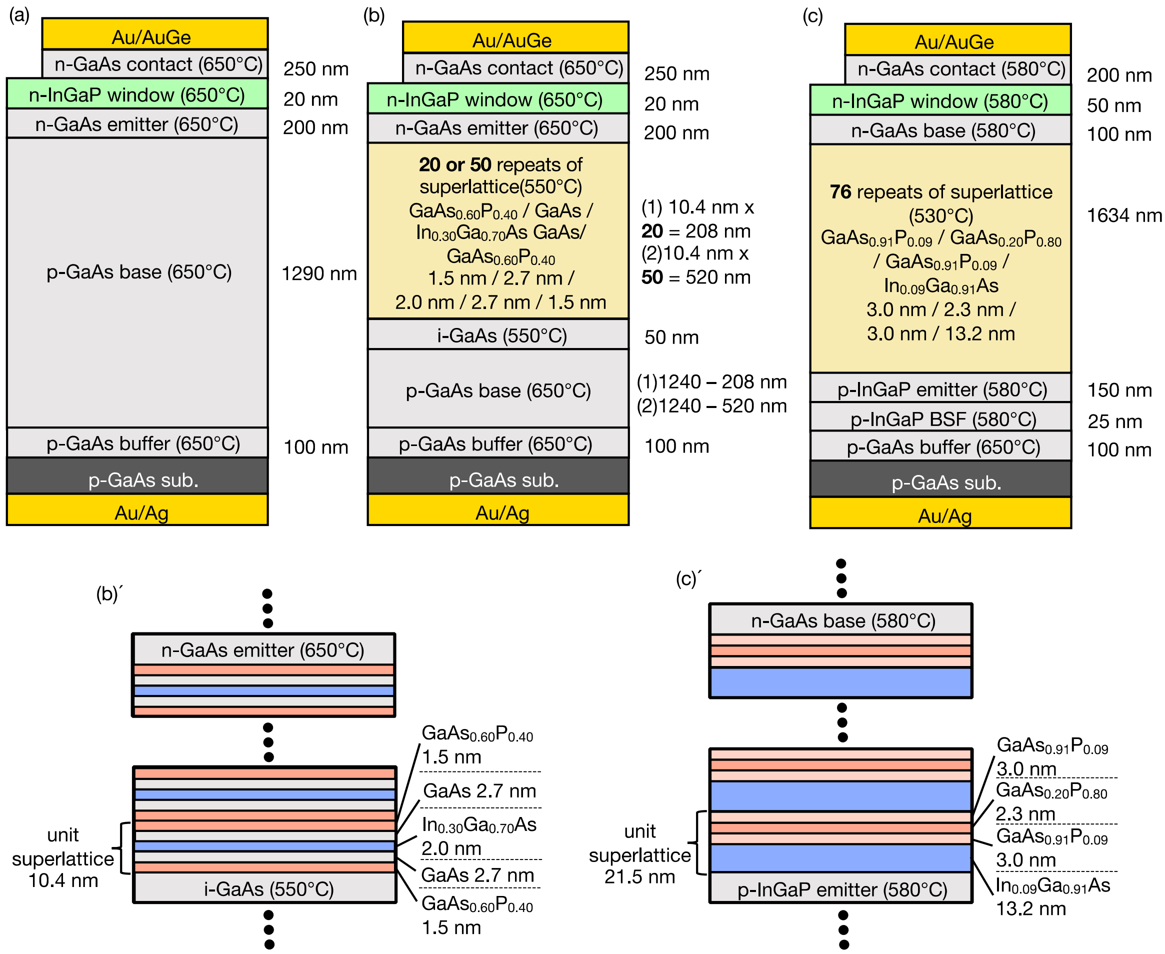

Fig. 4. Schematic of the sample structure. (a) GaAs reference sample, (b) conventional thin quantum well PSL and WoW samples, (c) Thick well PSL (TW-PSL) sample, (b)' structure of the superlattice of the samples shown in (b), and (c)' structure of the superlattice of the sample shown in (c). In (a), (b), and (c), the growth temperature of each layer is written in brackets.

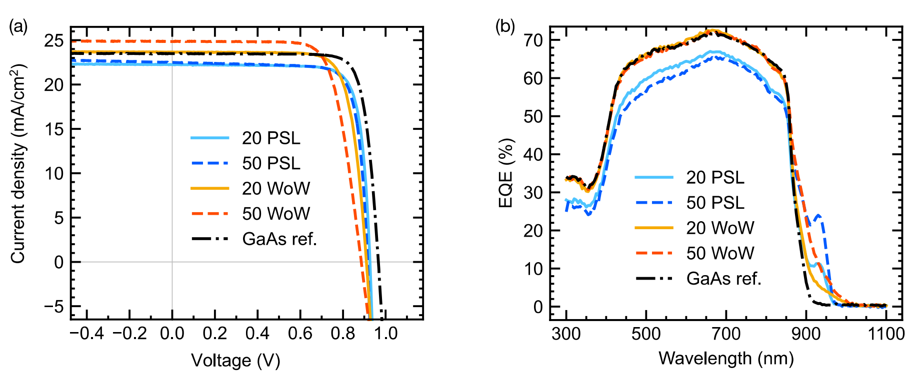

Fig. 5. (a) *J–V* characteristics, and (b) EQE of each sample.

Table 1
J–V characteristics of each solar cell.

Labels	GaAs ref.	20 PSL	50 PSL	20 WoW	50 WoW	TW-PSL
J_{SC} (mA/cm^2)	23.47	22.20	22.50	23.65	24.87	20.67
V_{OC} (V)	0.966	0.927	0.916	0.910	0.886	0.970
FF	0.814	0.813	0.803	0.779	0.736	0.798
Efficiency (%)	18.46	16.73	16.55	16.77	16.22	16.00

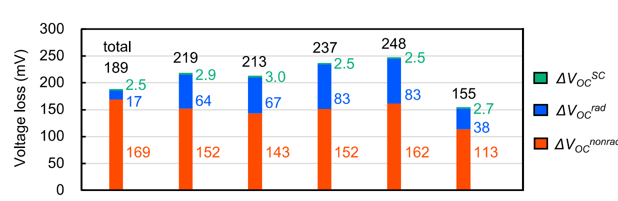

Fig. 6. Voltage-loss analysis of each sample.

the TW-PSL are discussed hereafter. In quantum-structured solar cells, the significant increase in voltage loss is attributed to radiative recombination. The radiative recombination voltage loss occurs when the EQE tail at the long-wavelength region gradually declines. A simple simulation was performed to elucidate the effect of the EQE tail. We assumed a double-step EQE model, as shown in Fig. 7(a). We calculated the radiative recombination voltage loss based on this EQE profile while varying the EQE value within 800–900 nm from 0 to 1. The resulting radiative recombination voltage loss is shown in Fig. 7(b).

In instances of low EQE, the voltage loss increased sharply owing to radiative recombination. This can be attributed to the fact that a low EQE in the long-wavelength region decreases the threshold voltage of the diode, despite its minimal contribution to enhancing the photocurrent. Similar to LEDs, solar cells composed of materials with small bandgaps possess a low threshold voltage and commence luminescence at lower voltages. Mitigating the radiative recombination voltage losses necessitates the steepening of the tail of the low EQE. Reducing voltage losses in solar cells requires addressing both radiative and nonradiative recombination voltage losses, while voltage losses resulting from short-circuit currents can be disregarded owing to their minimal absolute values and negligible variations across samples.

Through our analysis of voltage loss, we designed a novel quantum structure, shown in Fig. 4(c). We focused on the suppression of the voltage loss caused by radiative recombination. The gradually declining tails of the EQEs are the primary cause of voltage loss owing to radiative recombination. Therefore, we designed the EQEs to prevent the formation of these tails. The thickness of the quantum structures was initially considered. The change in the effective bandgap within a quantum well as a function of the thickness of the quantum structure is shown in Fig. 8. In this simulation, we assumed $GaAs_{0.200}P_{0.800}/In_{0.085}Ga_{0.915}As$ superlattices. The thickness of GaAsP was fixed to 3 nm, and the thickness of InGaAs was changed from 2 nm to 16 nm. As the thickness of the quantum structure increased, the rate of change in the bandgap became more gradual.

For example, consider a 4-nm-thick quantum well. In the event of a thickness deviation of ±2 nm in the quantum well layer, the corresponding energy level deviation would amount to 51 meV. However, if the same thickness deviation occurs in a 12-nm-thick quantum well, the energy-level deviation decreases to 18 meV. Because such energy-level fluctuations contribute to the EQE tail, increasing the thickness of the quantum well layer is advantageous to enhance its robustness against thickness fluctuations. During the growth of quantum structures using MOVPE, thickness fluctuations are inevitable, resulting in a small variance in the quantum well layer thickness. However, thickening the barrier layer introduces two issues.

1) Carrier transport: Quantum-structured solar cells rely on carrier transport through quantum tunneling and thermal escape, both of which are crucial. A thick barrier layer exponentially decreases the probability of carrier tunneling and degrades carrier transport.

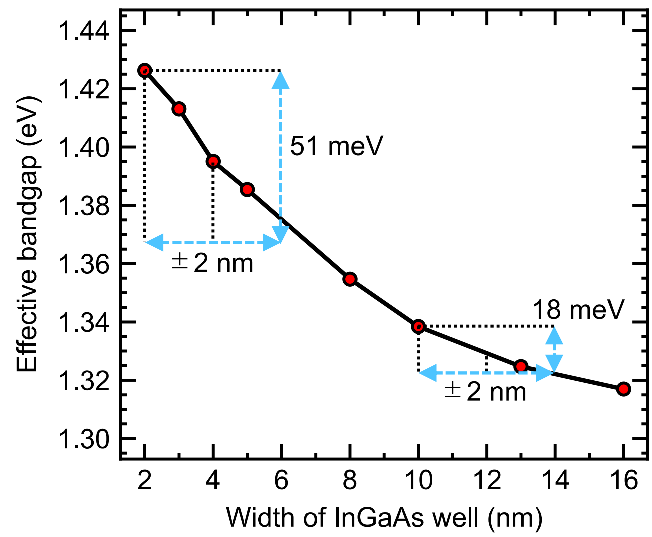

Fig. 8. Relationship between the effective bandgap and width of InGaAs quantum well. The red markers are simulated points.

2) Light absorption: Light absorption is sensitive to the thickness of the quantum well layer. In the case of a thick barrier layer, the proportion of the light-absorbing narrow-bandgap quantum well layer to the total thickness of the quantum structure decreases. For example, we consider two strain-balanced quantum structures:
   (a) $In_{0.1}Ga_{0.9}As/GaAs_{0.9}P_{0.1}$ (10 nm/20 nm), and
   (b) $In_{0.1}Ga_{0.9}As/GaAs_{0.2}P_{0.8}$ (10 nm/2.5 nm).
   In these cases, when fabricating a sample containing 100 repeats of quantum structures, the total thickness of the quantum structure in (b) would be 1.25 μm, whereas in (a), it would be 3 μm. This discrepancy poses disadvantages in terms of costs, such as those associated with solar cell fabrication time and raw material utilization. Furthermore, a thicker overall quantum structure results in a weaker internal electric field, which hinders carrier extraction. The electric field, determined from dV/dx (V and x are the voltage and position, respectively), weakens as the undoped quantum structure thickens. Additionally, creating a perfect undoped layer, that is the i-layer, is challenging, and the quantum structures in the i-region are often doped with background impurities, resulting in the p-type quantum structures. These impurity-induced carriers can be used to screen the electric fields. The thicker the quantum structure, the stronger the screening effect, making it more difficult to extract carriers. Therefore, a barrier layer must be thin to prevent the overall thickness of the quantum structure from becoming excessively large while ensuring high light absorption.

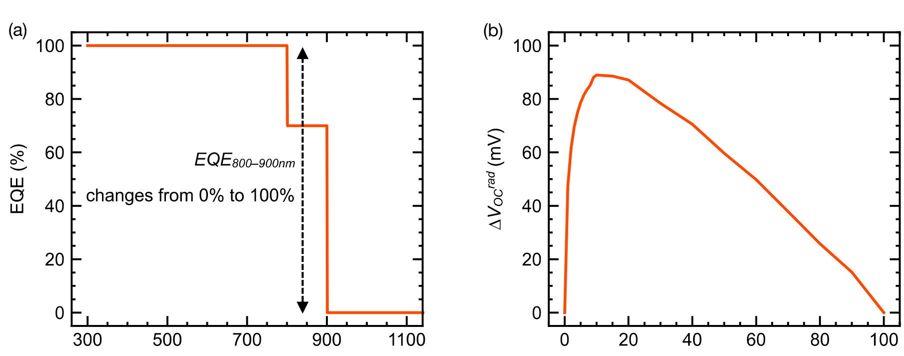

Fig. 7. (a) Assumed EQE for the simulation, and (b) simulation results of radiative recombination voltage loss.

For the aforementioned reasons 1) and 2), the barrier layer must be thin. In this study, we designed and fabricated a quantum structure based on these principles. The structural configuration is shown in Fig. 4 (c)'. To achieve thin barrier layers, we employed a GaAsP barrier layer with an extremely high phosphorous content of 80 %. However, this raised concerns regarding the potential formation of a substantial number of interface defects. To circumvent this problem, we inserted a GaAsP intermediate layer with low phosphorous content. Although the composition and thickness of the GaAsP intermediate layer and other layers can be optimized, this study primarily aimed to validate our proposed design of the quantum structure based on insights from the voltage-loss analysis.

The results of the *J–V* characteristics and EQE are shown in Fig. 9 and Table 1, respectively, and the voltage-loss analysis of the fabricated quantum-structured solar cells is presented in Fig. 6. First, we discuss the EQE results. The maximum EQE was 70 % at approximately 700 nm, which is sufficiently high. If the carrier mobility in the quantum structure is low, the overall EQE should decrease, as observed in the results of the conventional PSL samples. Thus, we believe that the carrier mobility in the TW-PSL is suitably high. The decrease in the short-wavelength side of the EQE of the TW-PSL sample may be attributed to suboptimal passivation by the window layer, resulting in more frequent recombination on the surface of the sample. Hence, the performance of the quantum structure is probably not compromised. Due to a low EQE at the short-wavelength side of the EQE profile, the TW-PSL sample showed the lowest short-circuit current density and power conversion efficiency among all the samples shown in Table 1. Although the growth condition of the window layer of the TW-PSL sample was not optimized, the TW-PSL sample was still superior to a conventional quantum structure considering its application to the middle cell of the triple-junction solar cells. Short-wavelength light is absorbed by the top InGaP (1.86 eV = 667 nm) subcell in triple-junction solar cells. Here, we assume that light with wavelengths longer than 667 nm enters the middle cell, whereas wavelengths shorter than 667 nm are cut off. We calculated the short-circuit current density of the GaAs ref., 50 WoW, and TW-PSL samples under this assumption, and the results were $J_{SC}^{\lambda>667nm} = 9.42$, 10.09, and 10.14 mA/cm^2, respectively. The $J_{SC}^{\lambda>667nm}$ value of the TW-PSL sample was the highest. The TW-PSL sample exhibited high current density and open-circuit voltage, and represents a superior quantum structure for fabricating the middle cell.

Compared with conventional quantum structures, the TW-PSL solar cell effectively halved the voltage loss owing to radiative recombination. This outcome validates the effectiveness of the proposed design strategy. By increasing the thickness of the quantum well layers, we successfully suppressed the gradually declining tail of the EQE, typically induced by the thickness fluctuations of the well layer; the thickness fluctuations in our samples resulted from indium carry over and As/P interdiffusion. The optimization of the quantum structure resulted in a steeper EQE edge and decreased voltage loss owing to radiative recombination.

We successfully mitigated the voltage loss caused by nonradiative recombination by reducing the indium content of the InGaAs layer. In our reactor, growing InGaAs epitaxially became challenging when the indium content exceeded 30 %. This difficulty may be attributed to the crystal strain experienced by InGaAs grown on GaAs, which can lead to indium segregation and formation of crystal defects [24]. By decreasing the indium content, we minimized the likelihood of segregation, and consequently, the occurrence of crystal defects. Another factor that contributes to the lower nonradiative recombination voltage loss in the TW-PSL sample is its GaAsP intermediate layer with a low phosphorous content. This layer helps mitigate the formation of interface crystal defects. Despite a GaAsP barrier layer with a high phosphorous content of 80 %, our voltage-loss analysis revealed no significant presence of interface defects.

Conventional quantum-structured solar cells have large radiative recombination voltage losses. Our proposed voltage-loss analysis technique is suitable for quantitative evaluation of solar cell voltage losses and their minimization. We successfully designed, fabricated, and demonstrated a novel high-performance quantum-structured solar cell. This study reveals the necessity of enhancing the steepness of the EQE edge and decreasing the occurrence of crystal defects for minimizing voltage loss owing to radiative and nonradiative recombination.

5. Conclusion

In this study, we introduced an innovative methodology to analyze voltage losses in solar cells. Conventional methodologies faced problems in evaluating the voltage losses derived from materials with accuracy, particularly when dealing with solar cells that have an unconventional EQE spectrum. Our proposed method addresses this challenge by normalizing the EQE by its maximum value, implementing suitable noise processing, and calculating the bandgap based on the generated photocurrent. We formulated a method to calculate the bandgap of solar cells fabricated using multiple materials. This method is based on the detailed balance theory and the equivalent photocurrent method.

We fabricated GaAs solar cells with InGaAs/GaAsP strain-balanced quantum structures and analyzed their voltage losses. Our findings demonstrate the significance of decreasing the voltage loss owing to radiative recombination. We observed that the radiative recombination voltage loss was caused by a gradually declining tail observed near the absorption edge of the EQE. Therefore, we enhanced the steepness of the EQE absorption edge by designing a novel quantum structure with thick quantum wells. Consequently, the radiative recombination voltage loss decreased from 80 mV for the WoW sample to 40 mV for the TW-PSL sample.

This study not only contributes to the assessment of the theoretical limitations of solar cells but also aids in the development of next-

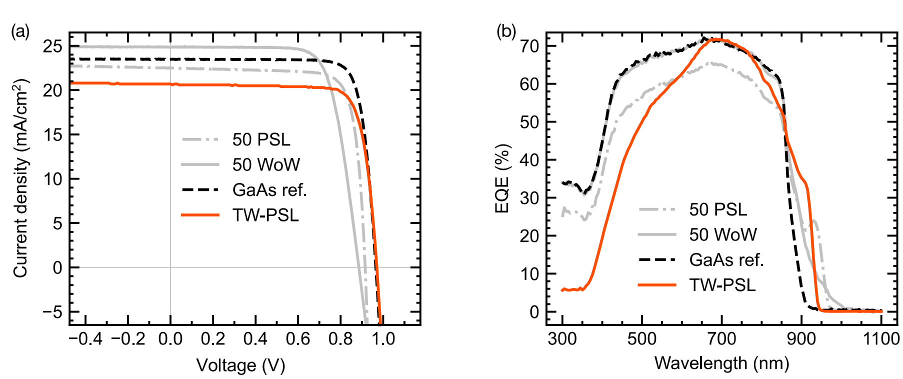

Fig. 9. (a) *J–V* characteristics, and (b) EQE of TW-PSL.

generation solar cells. The proposed methodology holds considerable promise as a practical tool for designing and evaluating next-generation solar cells, and it can be applied to semiconductors other than III-V compound semiconductors, such as perovskite, nanocolloidal quantum dot, chalcopyrite, and kesterite solar cells. Our proposed method enables the quantitative analysis of both radiative and nonradiative recombination voltage loss. Although we should reduce both radiative and nonradiative recombination voltage loss, we can decide which voltage-loss reduction we should intensively work on based on the voltage-loss analysis.

CRediT authorship contribution statement

Meita Asami: Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Maui Hino:** Writing – review & editing, Validation, Data curation. **Gan Li:** Writing – review & editing, Validation, Data curation. **Kentaroh Watanabe:** Writing – review & editing, Supervision, Resources. **Yoshiaki Nakano:** Writing – review & editing, Supervision, Resources. **Masakazu Sugiyama:** Writing – review & editing, Validation, Supervision, Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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